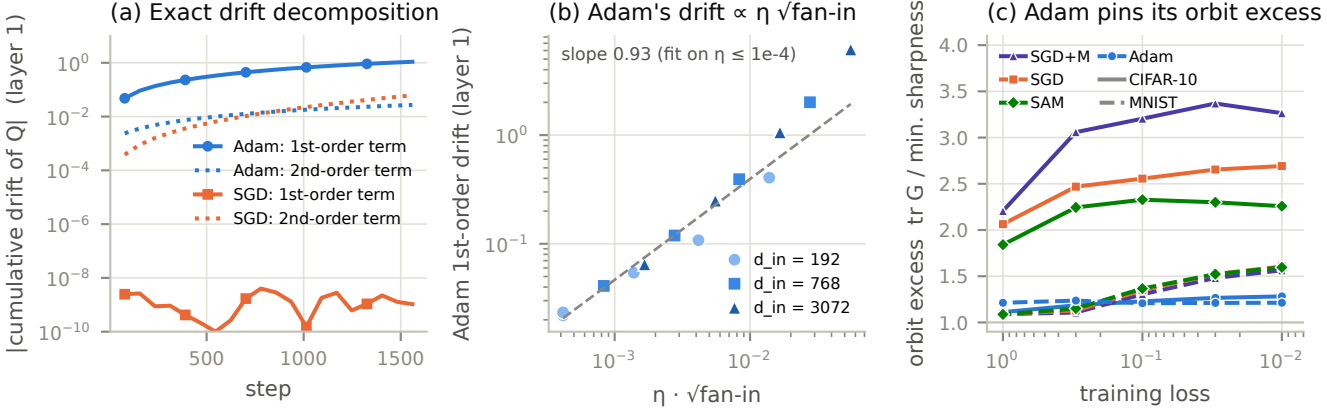
(a) Exact drift decomposition
|cumulative drift of Q| (layer 1)
step
Adam: 1st-order term
Adam: 2nd-order term
SGD: 1st-order term
SGD: 2nd-order term
(b) Adam's drift ∝ η √fan-in
Adam 1st-order drift (layer 1)
η · √fan-in
slope 0.93 (fit on η ≤ 1e-4)
d_in = 192
d_in = 768
d_in = 3072
(c) Adam pins its orbit excess
orbit excess tr G / min. sharpness
training loss
SGD+M
Adam
SGD
CIFAR-10
SAM
MNIST